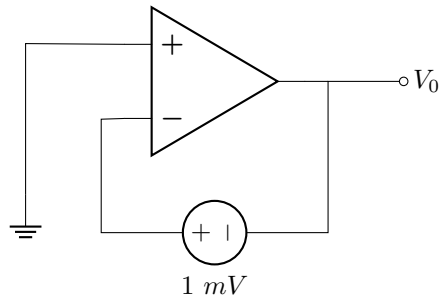
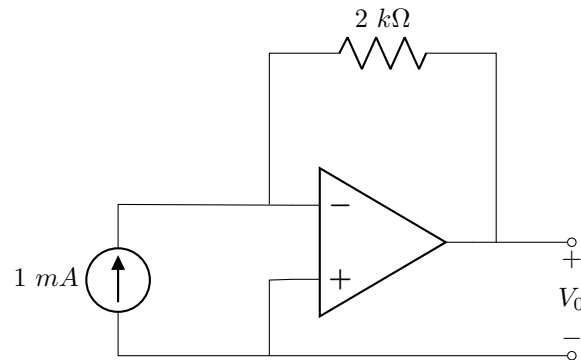
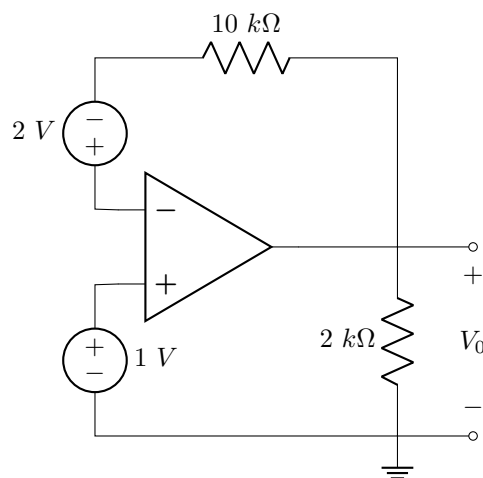


**Tutorial 6**

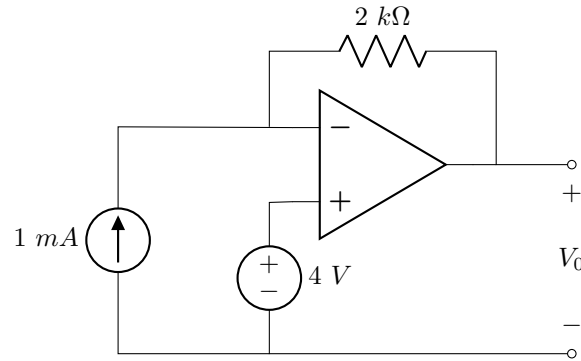
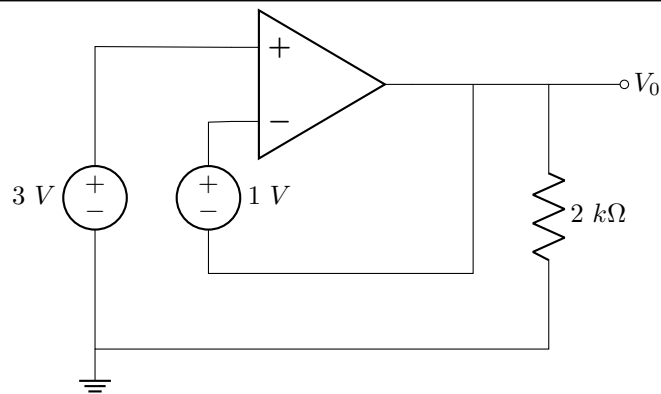
Question 5.6.

**01** Determine  $V_0$  if this a 741 Opamp (it is not ideal) [1]

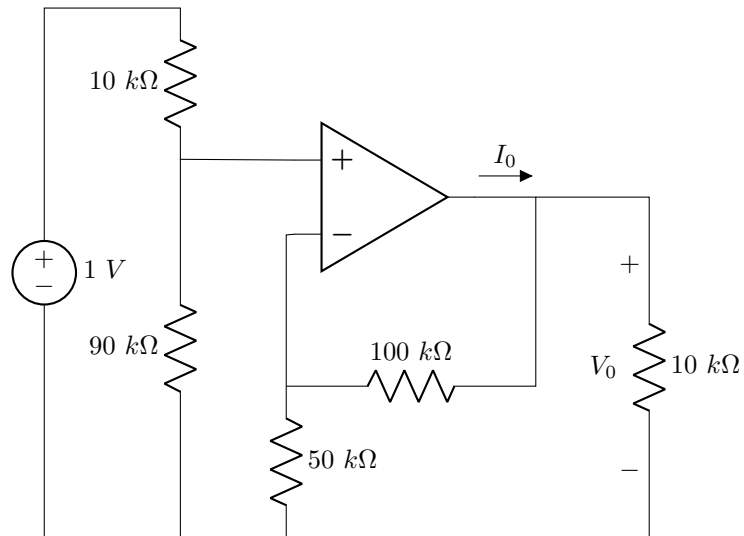
## Question 5.8.

**02** Determine  $V_0$  [1]**03** Determine  $V_0$  [1]

## Question 5.9.

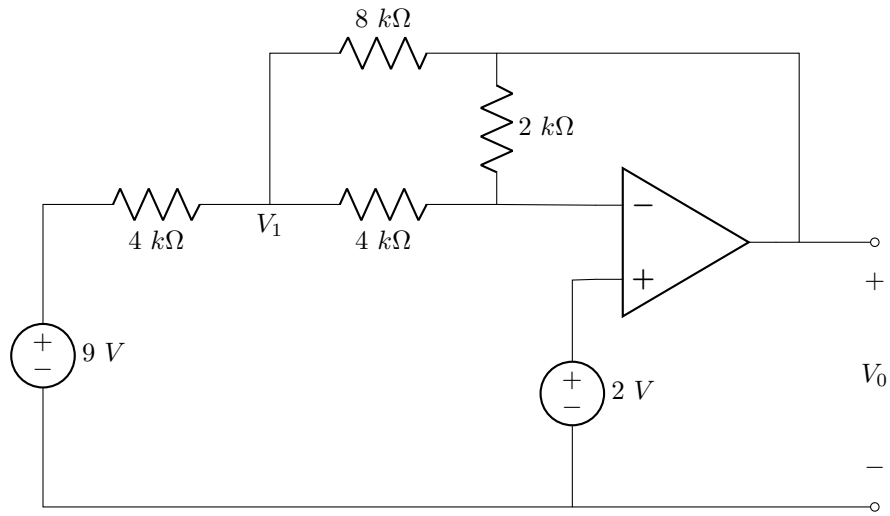
**04** Determine  $V_0$  [1]**05** Determine  $V_0$  [1]

## Question 5.13.

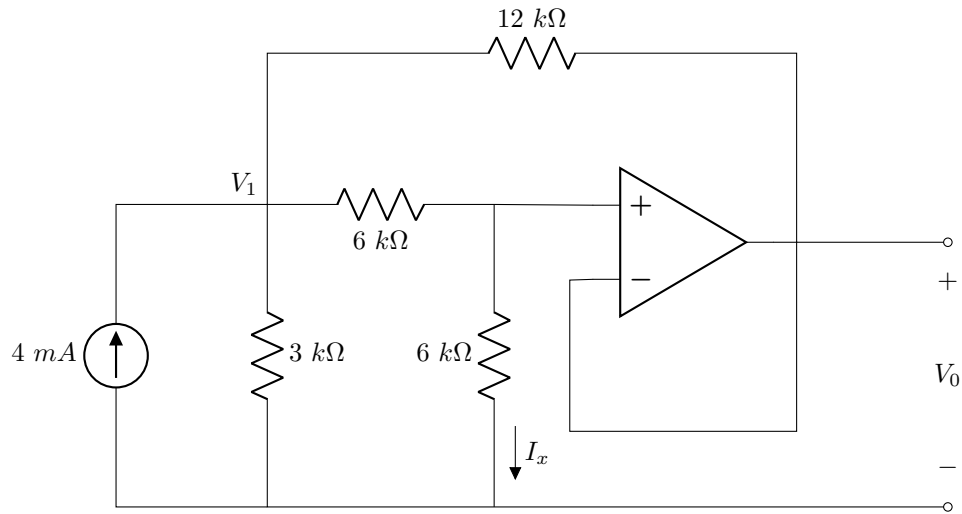


**06** Determine  $I_0$  [2]

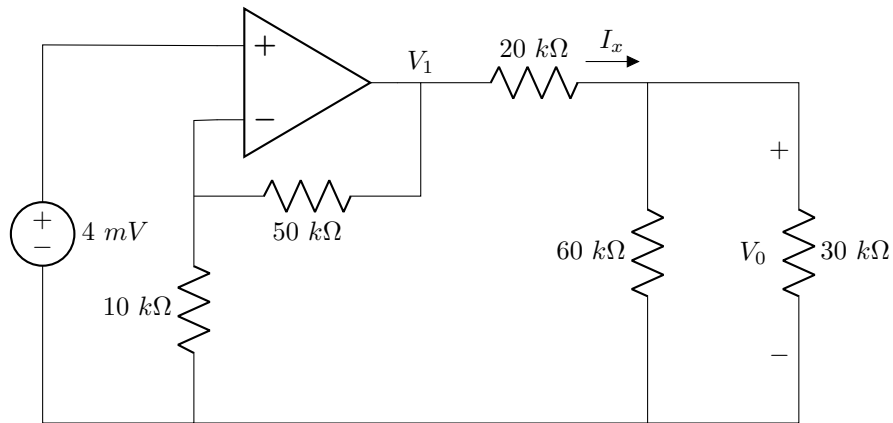
## Question 5.20.

**07** Determine  $V_0$  [2]

## Question 5.31.

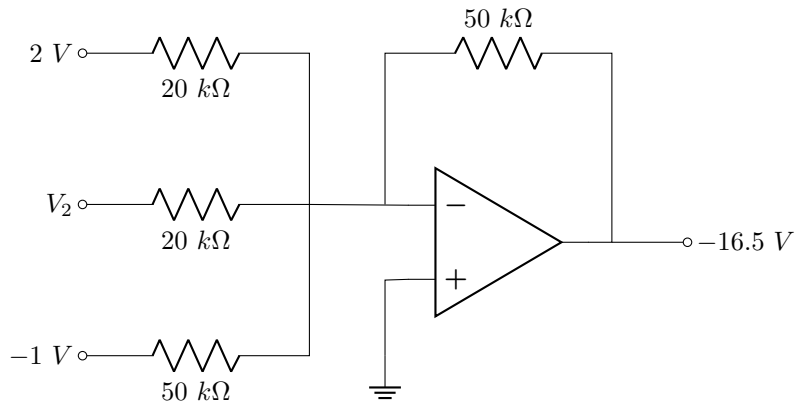
**08** Determine  $V_0$  [2]**09** Determine  $I_x$  [2]

Question 5.32.



**10** Determine  $I_x$  and  $V_0$  [4]

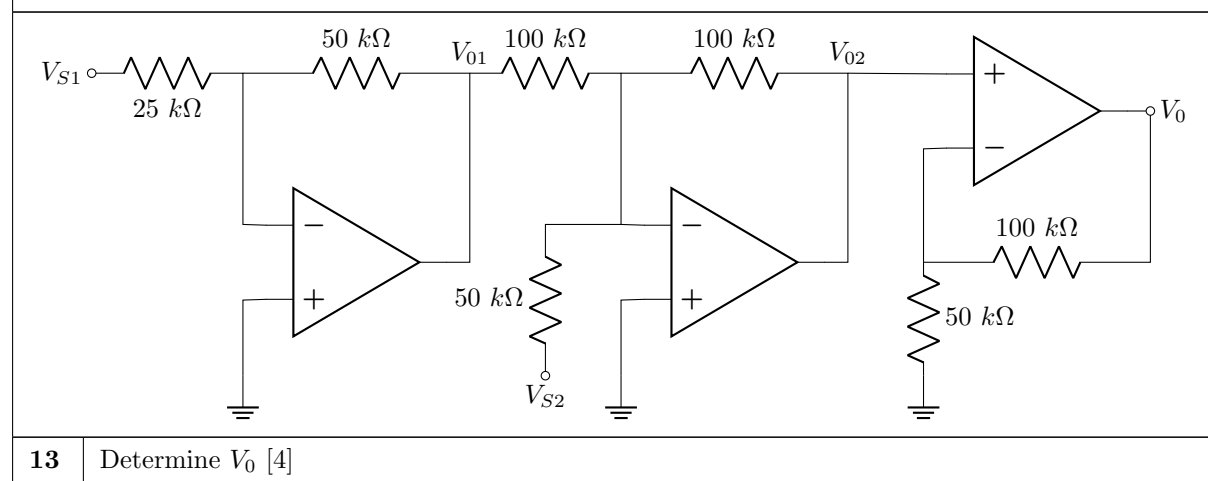
## Question 5.39.



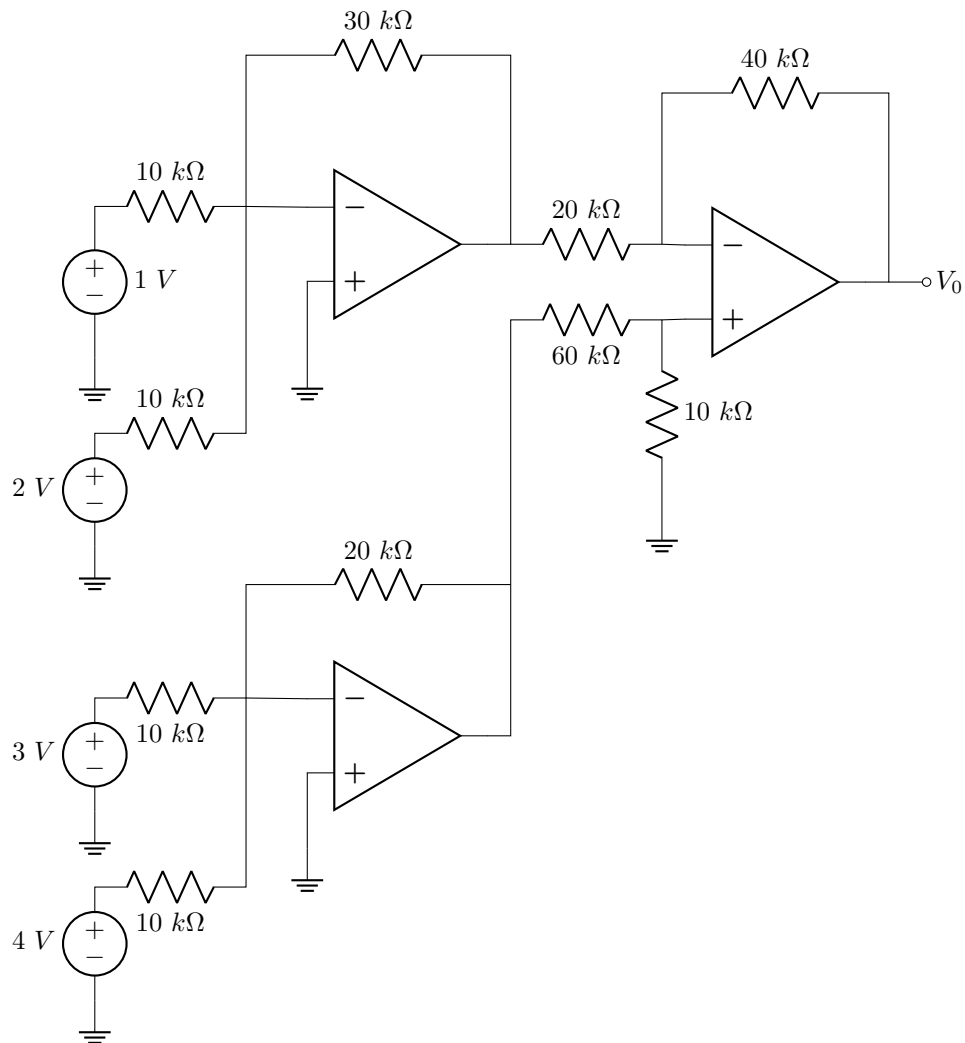
- 11 Determine  $V_{TH}$  and  $R_{TH}$  across  $a - b$  [4]

|                |                                                                                       |
|----------------|---------------------------------------------------------------------------------------|
| Question 5.45. |                                                                                       |
| <b>12</b>      | Design an op-amp circuit with the following I-O relationship: $V_0 = 3V_1 - 2V_2$ [4] |

Question 5.57.



Question 5.70.

**14** Determine  $V_0$  [4]